

PROFI BUS

Architecture, determinism, and the road to Industrial Ethernet

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ANATOMY OF AN INDUSTRIAL PROTOCOL

Determinism

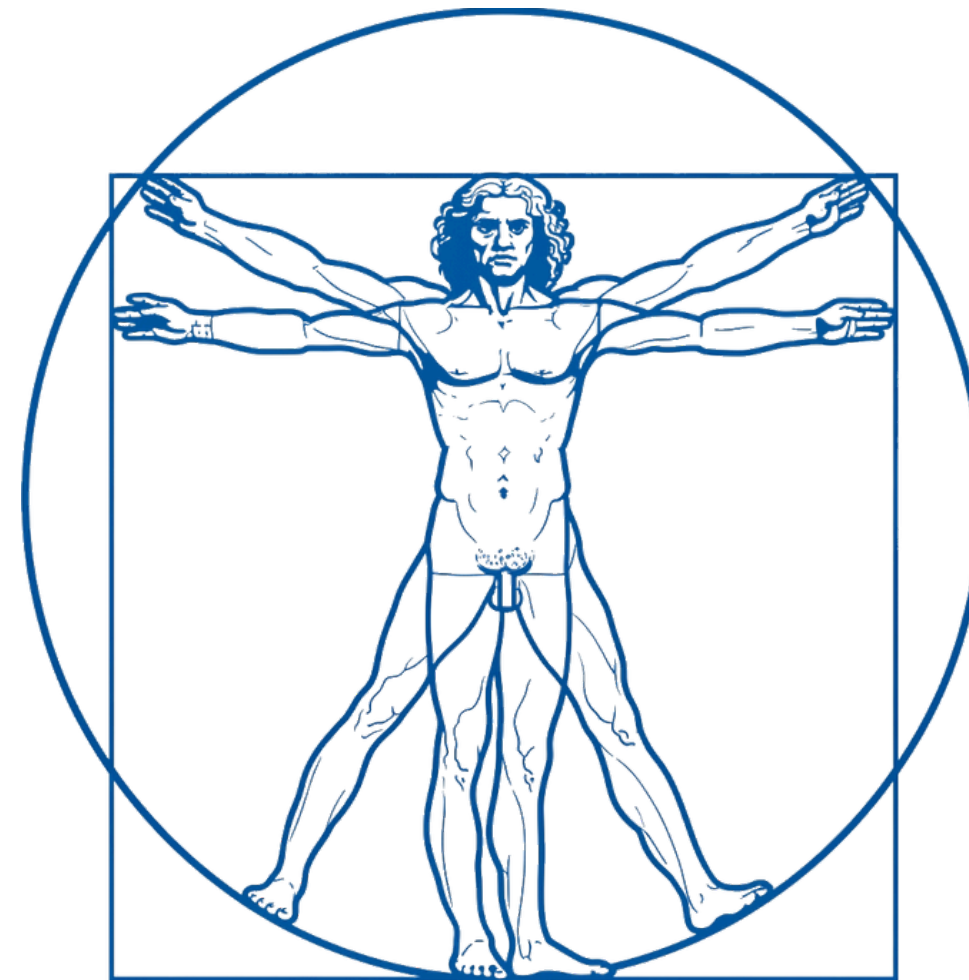
Bus access and delivery bounded in time.

Reliability

Predictable behaviour under electrical noise

Diagnostics

Devices report module- and channel-level faults



4

Interoperability

Lets multi-vendor devices coexist.

5

Efficiency

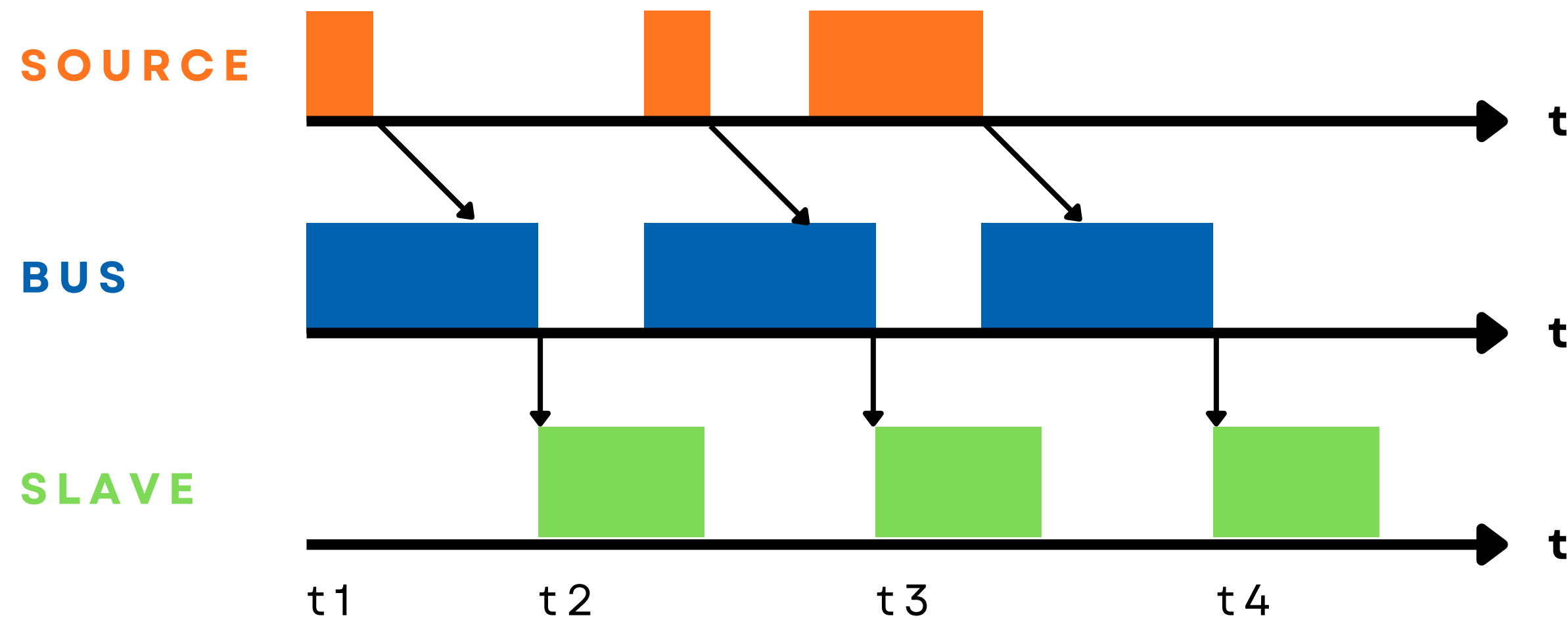
Compact telegrams move whole I/O images

6

Safety

The physical layer must be intrinsically safe.

DETERMINISM

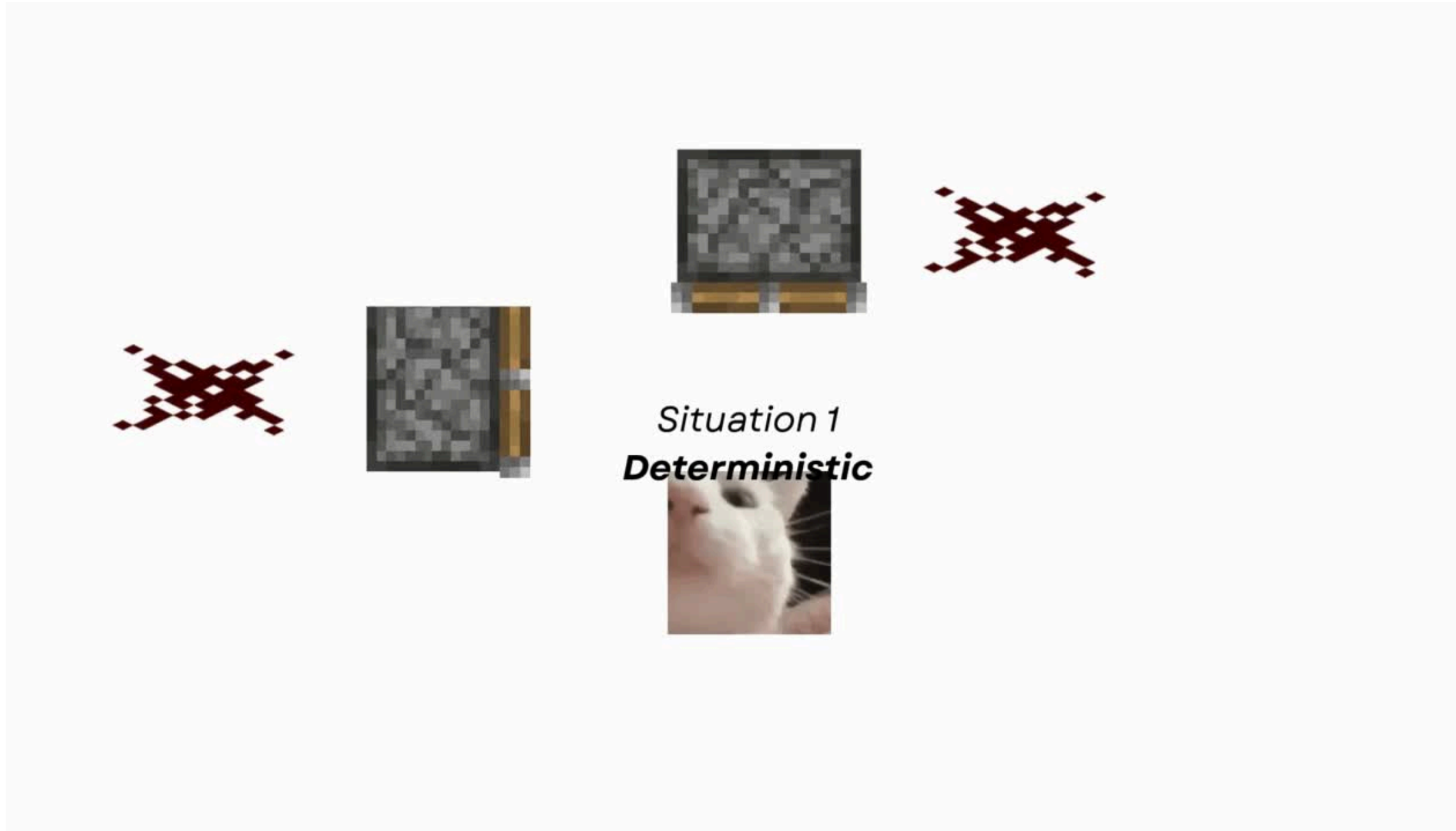


EXAMPLE

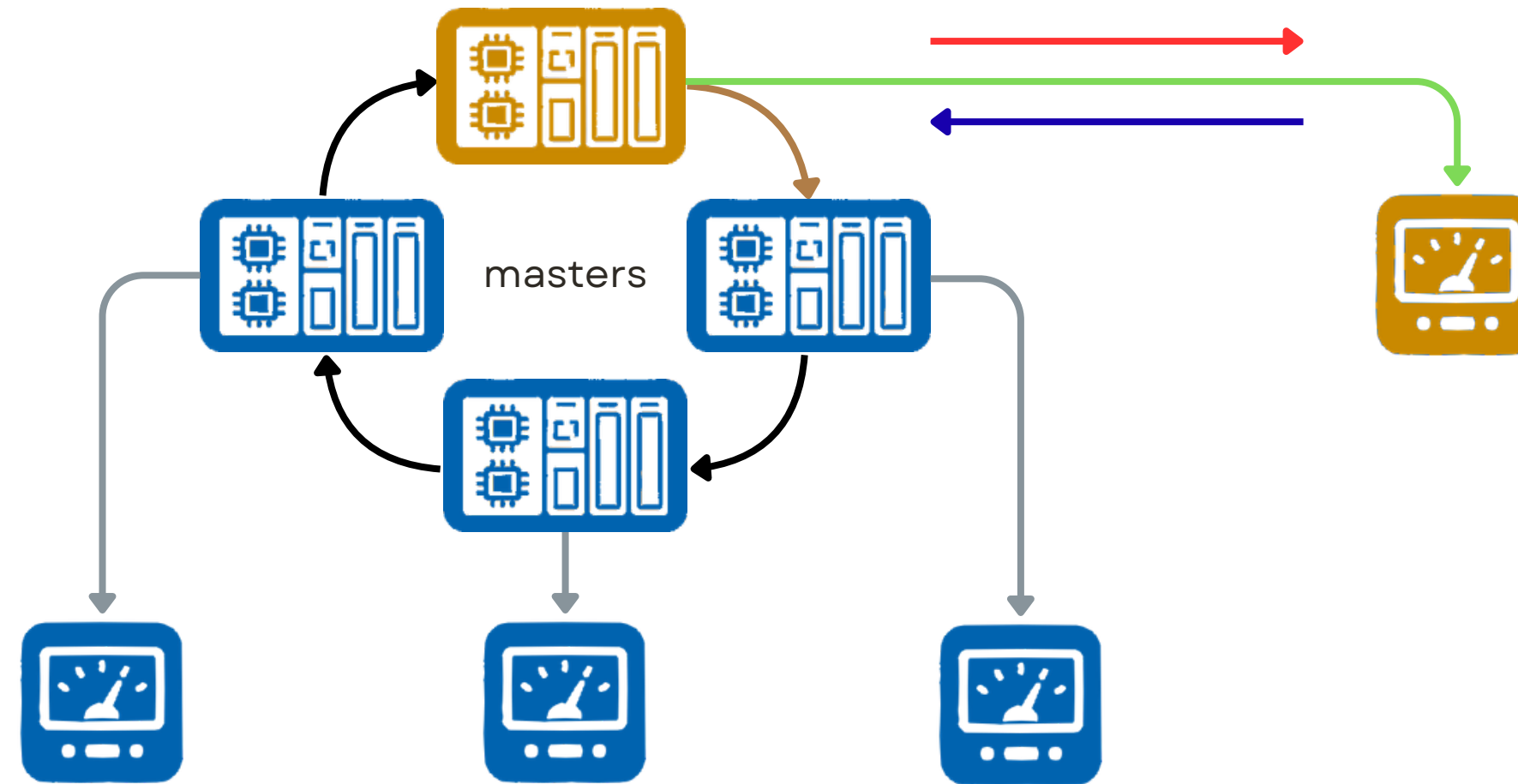


GOAL: An automated assembly that puts the hard-hat on the cat

EXAMPLE



How PROFIBUS Guarantees It

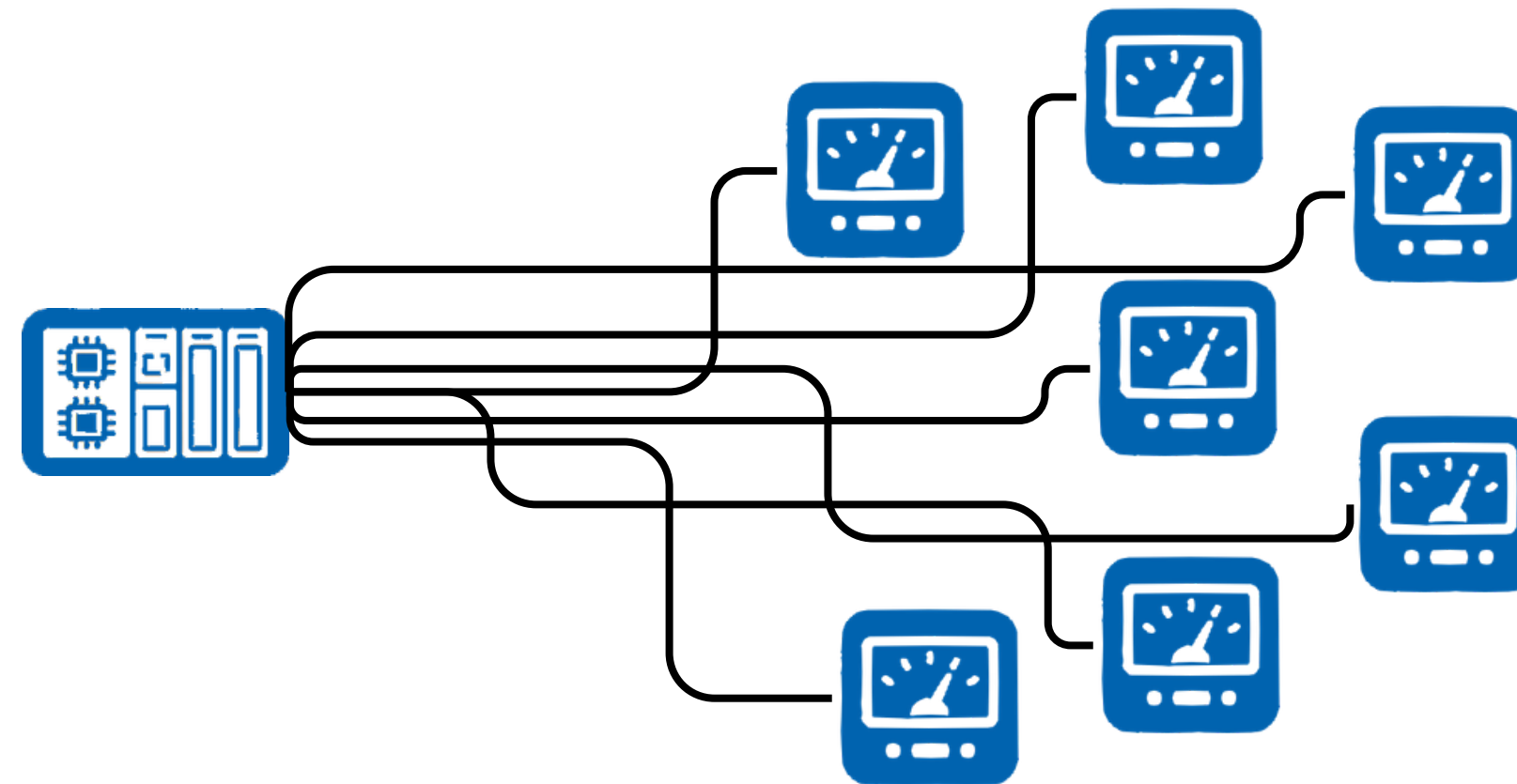


- Masters form a logical token ring where only the token-holder controls the bus
- That master polls its slaves request-response; a slave speaks only when addressed
- Target Token Rotation Time (TTR) bounds the worst-case wait for every station

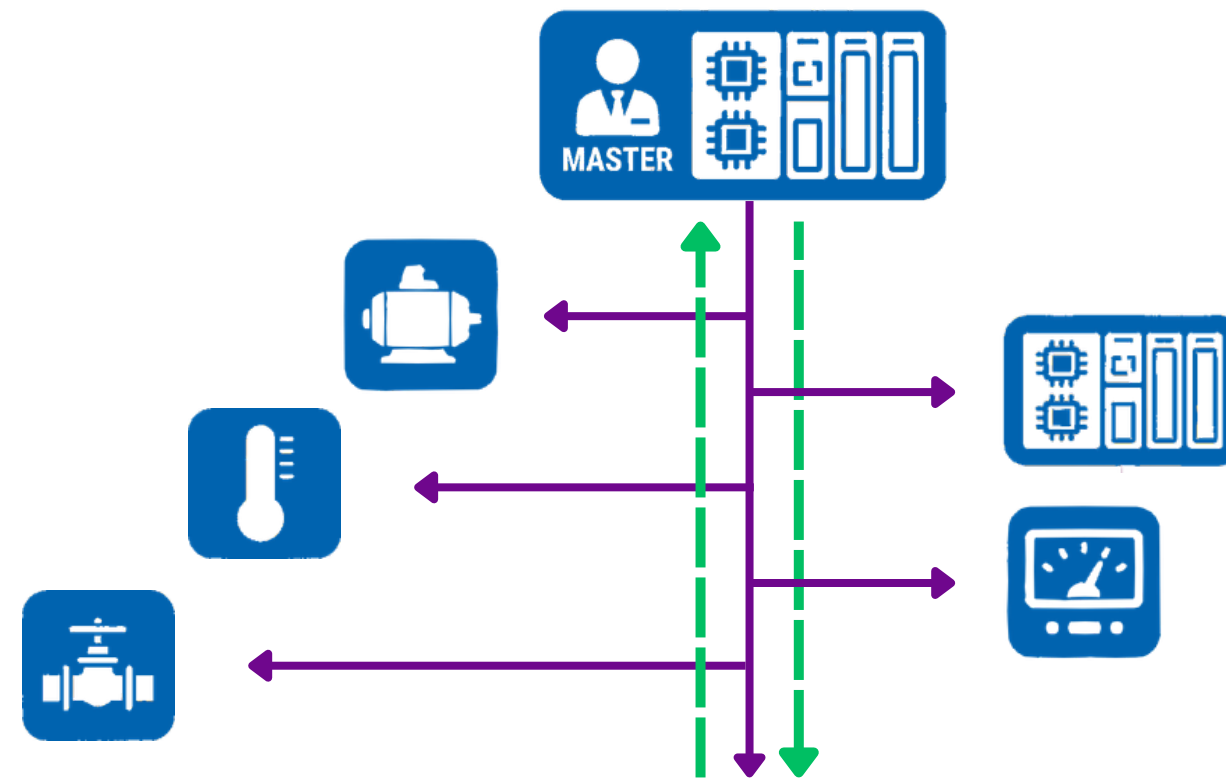
Before PROFIBUS

The state of the art in the early 1980s

- Each sensor and actuator wired individually back to a central controller.
- Analog 4–20 mA signalling was unidirectional: no device diagnostics, no remote configuration.
- Installation was expensive, error-prone, and difficult to maintain as facilities grew



Hello, PROFIBUS!



- Many devices share a single serial cable
- Bidirectional exchange: data, parameters and diagnostics flow both ways
- Control decentralised toward intelligent field devices

Evolution & History

1987



Open standard

German firms & institutes launch a joint open-fieldbus project

1989



PROFIBUS-FMS

Cell-level messaging between controllers; powerful but heavy and slow

1993



PROFIBUS-DP

Fast cyclic I/O to remote peripherals which became the industry standard

1990s



PROFIBUS-PA

Intrinsically safe, bus-powered variant for process & hazardous areas

DP and PA dominate today; FMS is effectively obsolete.

Adoption & Utility

A massive install base, split cleanly by what each variant is good at.

PROFIBUS-DP

factory & discrete automation

- Discrete manufacturing and assembly lines
- Motion control and drives, where reaction time is critical
- High-speed cyclic I/O at up to 12 Mbps

PROFIBUS-PA

process & hazardous-area automation

- Chemicals, oil & gas, water and wastewater
- Explosive atmospheres needing intrinsic safety
- Continuous variables: flow, level, pressure, temperature

Architecture & Communication Model

Key Features



Master-slave
communication model



Multi-vendor interoperability
via GSD & EDDL



Determinism via Target
Token Rotation Time

Three variants:

- **DP** decentralized control & I/O communication
- **PA** process automation
- **FMS** peer-to-peer communication (now largely obsolete)

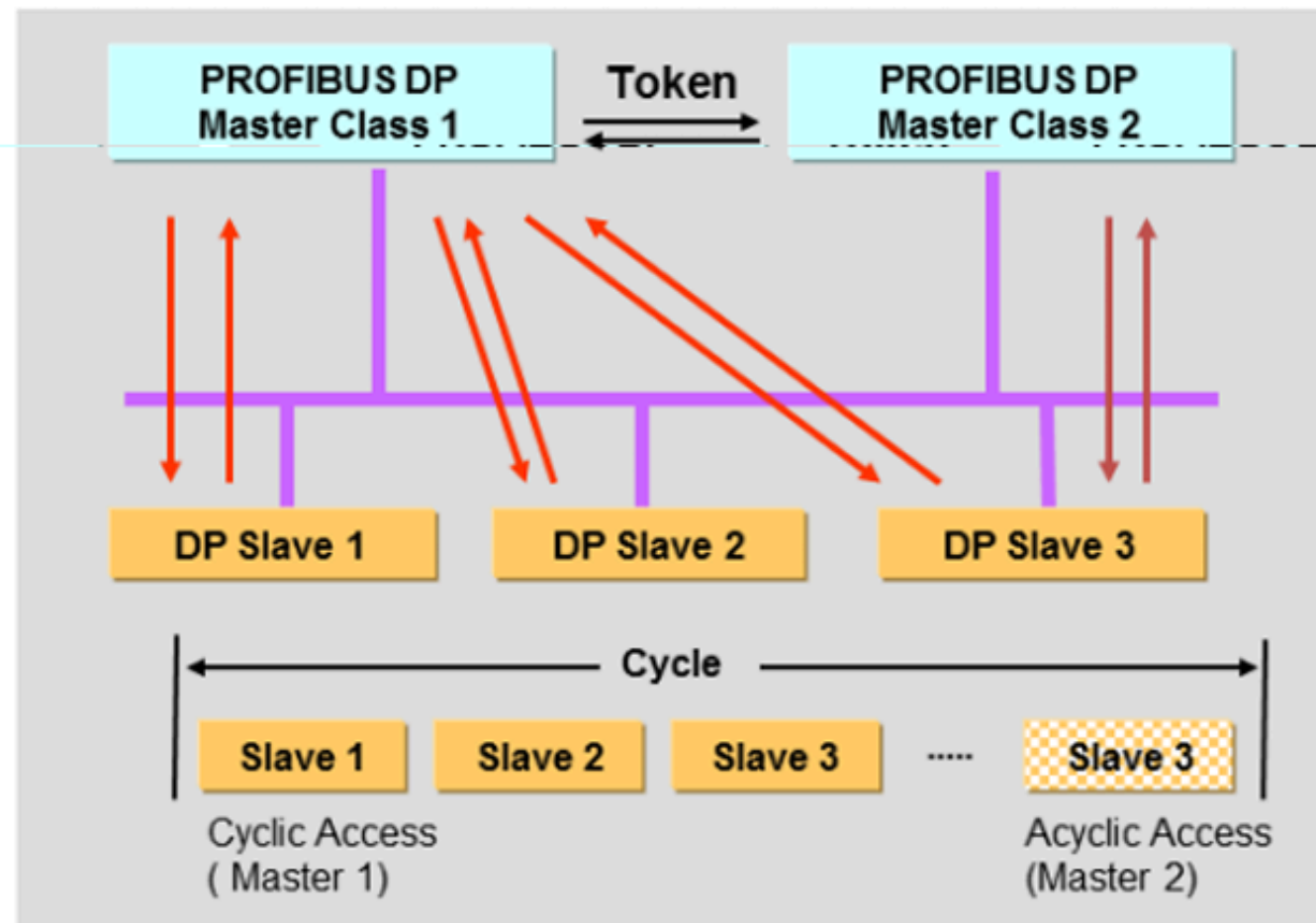
PROFIBUS ARCHITECTURE & OPERATION

OSI defines communication across 7 layers; PROFIBUS deliberately implements only L1, L2 & L7

- **L1 Physical:** RS-485 copper, fibre-optic, intrinsic-safety variants include RS485-IS and MBP-IS.
- **L2 Data Link (FDL):** bus access = master-slave + token
- **L7 Application:** DP protocol / FMS + LLI + FMA7

7	Application	DP protocol (V0/V1/V2) · FMS + LLI + FMA7
6	Presentation	<i>not used</i>
5	Session	<i>not used</i>
4	Transport	<i>not used</i>
3	Network	<i>not used</i>
2	Data Link	Fieldbus Data Link (FDL): master-slave + token
1	Physical	RS-485 · MBP · fibre-optic · intrinsically safe

Bus Access: Hybrid Master-Slave + Token



- Token-passing among masters + master-slave polling of slaves
- Every station gets bus access within a fixed time
- Strict request-response: a slave replies only when addressed
- Timing: $TTR_{target} \cdot TRR_{actual} \cdot TTH_{hold} = TTR - TRR$
- High-priority telegrams first; low-priority only if time remains

Communication Services & PROFIBUS-DP

Cyclic

Time-critical process I/O. Executed every bus cycle, typically by a class 1 master.

12 Mbps

max data rate

244 bytes

per cycle → full I/O update in a single cycle

Acyclic

On-demand: configuration, parameterization, diagnostics, alarm handling.

Why split them?

Cyclic data stays predictable; acyclic tasks like diagnostics run only when needed and can take multiple cycles.

- Core variant; provides the fundamental functions the other versions build on
- Fast, decentralized control of PLCs, distributed I/O & drives
- Master-slave, cyclic: master sends outputs (setpoints) → slave returns inputs (measurements)
- Runs mainly on RS-485 (speed + noise immunity); also fibre-optic or wireless

Functional Levels: DP-V0 / V1 / V2

DP is layered into three performance levels each one builds on the last.

Level	Core function	Adds
DP-V0	Basic cyclic data exchange	GSD-based config; device-, module- & channel-specific diagnostics for fast fault localisation (e.g. "excess temperature", "short circuit on output"). Data format depends on device profile.
DP-V1	Acyclic communication	Parameterization, operation, monitoring, alarm handling; online access to bus nodes via engineering tools
DP-V2	Drive-control extensions	Slave-to-slave communication, cycle synchronization, time-stamping

Note: *process-automation field devices are typically DP-V1 slaves, so they can be parameterized acyclically.*

PROFIBUS-PA: Process Automation

- Intrinsically safe operation, long cable lengths, bus-powered field devices
- Power + communication on the same two-wire cable → lower installation cost
- PA profile standardizes device functions, parameters, diagnostics & cyclic data formats → interoperability + simpler engineering
- Physical layer = MBP (Manchester Coded, Bus Powered); single twisted-pair Type A cable

31.25 kbps

fixed rate · Manchester II · ± 9 mA

~1900 m

max segment length

32 / ≥ 9 V

devices per segment · ≥ 9 V DC, 10 mA each

PA: Integration & Hazardous Areas

PA is a subsystem of DP – needs a coupler or link to bridge segments:

Coupler	Link
Forwards signals; PA devices appear as DP stations. Fixes the DP baud rate.	Intelligent gateway with its own DP address. Allows flexible DP baud rates, more independent operation.

Into higher-level systems (e.g. PROFINET): the controller acts as gateway, connecting PA devices to Ethernet for centralized control

Process-data representation (PA profile):

- Analog → 4-byte IEEE float + 1 status byte
- Digital → 1 data byte + 1 status byte

Hazardous-area installation concepts:

FISCO	High-Power Trunk
Simplifies intrinsic-safety verification – certified components within predefined electrical limits	Separates intrinsically-safe spurs from a higher-power trunk cable

PROFIBUS-FMS

- The first variant, introduced in 1989
- Object-oriented protocol for cell-level communication between intelligent controllers
- Emphasizes functionality & acyclic exchange of complex data – not fast response
- Application layer = FMS + LLI; supports dynamically established and predefined communication relationships
- Relationships managed via a Communication Relationship List (CRL); application profiles ensure vendor interoperability
- High overhead, lower effective speed, reduced determinism, hence largely obsolete

The Three Variants

	PROFIBUS-DP	PROFIBUS-PA	PROFIBUS-FMS
Purpose	Fast decentralised I/O control	Process automation, hazardous areas	Cell-level messaging (obsolete)
Physical layer	RS-485 (fibre / wireless)	MBP, bus-powered 31.25 kbps	RS-485 / fibre
Exchange style	Fast cyclic + acyclic	Cyclic process data + acyclic	Acyclic, complex objects
Speed	up to 12 Mbps	fixed 31.25 kbps	low, high overhead
Status today	Dominant standard	Widely deployed in process	Effectively retired

Limitations & Improvements

Integration & Maintenance

Limitation

Large PROFIBUS systems with hundreds or thousands of devices become harder to integrate and maintain, especially as equipment ages.

Improvement

Use PROFIBUS/PROFINET gateways, unified engineering tools, IoT/cloud-based device-data collection, energy monitoring, and predictive maintenance.

Emerging Competition

Limitation

Ethernet-based protocols such as PROFINET, EtherCAT, and Modbus TCP offer higher data rates, richer topologies, and stronger data-processing capability.

Improvement

Ethernet-APL / 10BASE-T1L provides an Ethernet-based path over two-wire cabling with power and data delivery, suitable for harsh and intrinsically safe environments.

Technology Upgrading

Limitation

New industrial devices increasingly support newer protocols, while existing PROFIBUS systems must remain compatible or face costly upgrades.

Improvement

PROFINET can act as an integration platform, allowing existing PROFIBUS technology to be included in real-time Ethernet systems without full replacement.

Limitations & Improvements

Physical-layer Fragility

Limitation

RS-485 communication is sensitive to long cabling, missing terminators, electromagnetic interference, open circuits, and short circuits.

Improvement

Automated physical-layer diagnosis using artificial neural networks can classify signal waveforms and identify installation faults.

Wired-only Architecture

Limitation

PROFIBUS is fundamentally wired, which limits mobility and makes deployment harder in difficult-to-cable areas.

Improvement

A dual-link OPC UA gateway using fibre plus mobile data can send process data to a remote client while keeping the local real-time loop wired.

No built-in Cybersecurity

Limitation

PROFIBUS has no native authentication or encryption, so a rogue device with bus access can read telegrams, spoof traffic, map topology, or manipulate devices.

Improvement

DpGuard learns normal PROFIBUS-DP traffic behavior and detects anomalies, including fault-injection and semantic attacks, with microsecond-scale overhead.

Profibus vs Profinet

	<u>Profibus DP</u>	<u>Profinet</u>
Network Type	Serial fieldbus over RS-485	Industrial Ethernet
Speed	9.6 kbit/s to 12 Mbit/s	Higher Ethernet-based bandwidth
Communication Model	Master-slave polling	Switched Ethernet communication
Determinism	The master controls the polling sequence	Real-time options through PROFINET RT and IRT
Cycle Time	About 5 ms for 20 slaves at 1.5 Mbit/s	RT for general real-time tasks; IRT for stricter synchronization
Best Use	Existing fieldbus systems and legacy plants	New Ethernet-based automation and migration from PROFIBUS

Profibus vs EtherCAT

	<u>Profibus DP</u>	<u>EtherCAT</u>
Network Type	Serial fieldbus	Ethernet-based fieldbus
Frame handling	Master polls slaves cyclically	One Ethernet frame passes through all slaves
Cycle time	Configuration-dependent; examples in paper include 5 ms and 0.77 m	Designed for cycle times $\leq 100 \mu\text{s}$
Jitter	Predictable but not designed for sub-microsecond synchronization	Jitter $\leq 1 \mu\text{s}$; distributed clocks can synchronize very precisely
Scalability	Up to 126 stations total; 32 per RS-485 segment	Up to 65,535 nodes per network segment
Best Use	Factory automation, distributed I/O, legacy control	Robotics, CNC, servo drives, semiconductor equipment, multi-axis motion

Profibus vs Modbus vs EtherNet/IP

	<u>Profibus DP</u>	<u>Modbus TCP</u>	<u>EtherNET/IP</u>
Main Strength	Deterministic fieldbus communication	Simple, open, widely supported	Uses CIP on standard Ethernet infrastructure
Real-time Behavior	Predictable due to master-slave polling	Non-deterministic, it uses TCP/IP request-response communication	Soft real-time normally; hard real-time possible through CIP Motion
Best Application Area	Legacy automation, distributed I/O, reliable plant control	SCADA, monitoring, energy systems, legacy integration	North American automation, food and beverage, automotive assembly
Compared with PROFIBUS	More deterministic than Modbus TCP, but less flexible than Ethernet-based systems	Easier and simpler, but weaker for real-time control	More Ethernet-compatible, but standard EtherNet/IP depends on network load

Conclusion

PROFIBUS remains reliable and deterministic as master-slave + token give predictable exchange, and GSD/EDDL enable multi-vendor integration, but its limited bandwidth and absence of security make it a poor fit for high-speed, IT/OT-converged Industry 4.0 systems